

Assignment 1

Number System conversion 1's and 2's compliment

Q1. CONVERT BINARY TO DECIMAL

1. $(110101)_2$
2. $(101.1101)_2$
3. $(10101)_2$
4. $(1011.101)_2$
5. $(100101)_2$

Q2. CONVERT DECIMAL TO BINARY

1. $(0.78125)_{10}$
2. $(13.6875)_{10}$
3. $(192)_{10}$
4. $(653.625)_{10}$

Q3. CONVERT HEXADECIMAL TO DECIMAL

1. $(73D5)_{16}$
2. $(39.B8)_{16}$
3. $(129A.B86)_{16}$
4. $(39.C8)_{16}$

Q3. CONVERT HEXADECIMAL TO BINARY

- (a) $(3D59)_{16}$
- (b) $(27.A3C)_{16}$
- (c) $(129A.B86)_{16}$

Q4. CONVERT BINARY TO HEXADECIMAL

- (a) $(10110100101110)_2$
- (b) $(11100.1011011011)_2$

Q5. CONVERT BINARY TO OCTAL

- (a) $(1101101111)_2$
- (b) $(010101101)_2$

Q6. ADD THE FOLLOWING NUMBERS

1. $(100011)_2$ and $(101)_2$
2. $(101)_2$ and $(1011)_2$

Q7. EVALUATE THE FOLLOWING NUMBERS

1. $(10010)_2 - (11011)_2$
2. $(10101)_2 - (00110)_2$
3. $(11101)_2 - (1011)_2$

Q8. FIND 1's AND 2's COMPLIMENT

1. $(101101)_2$
2. $(111100001111)_2$
3. $(9090)_{10}$
4. $(1010101)_2$

Q9.PERFORM SUBTRACTION USING 1's AND 2's COMPLIMENT

1. $(11011)_2 - (10010)_2$
2. $(10001110)_2 - (11110000)_2$
3. $(10101)_2 - (00110)_2$